

Hackathon Rules and Criteria Compliance Review

1. In-Hackathon Development Requirement

The provided materials do not explicitly state “all work must be done on-site during the hackathon,” but the event’s format strongly implies it. The challenge problem and dataset are only **revealed at the end of the kickoff workshop on Friday night**, after which teams “get to work” on their solutions ¹. In other words, no team should have a completed solution before the hackathon begins, since the “*mystery data set*” isn’t released until kickoff ². Many hackathons operate under the understanding that projects are developed within the competition window, and this one is no exception. While we did not find a written rule outright banning pre-existing code, it’s clear the organizers expect you to **build your solution during the Jan. 23-25 event** using the newly provided problem and data. Teams also had to meet certain composition rules (each team needed at least one UT PGE student and one undergraduate) ³, emphasizing that the hackathon is a student-focused, collaborative event.

Potential Conflict: Your current approach involves a fully developed pipeline built *before* the event using last year’s data. This could conflict with the **spirit of the competition**. Even if not explicitly forbidden in writing, showing up with a pre-built solution may draw scrutiny. Judges highly value originality and creativity (discussed below), so a project that appears pre-made or recycled could score poorly on those fronts. To stay safe, plan to **do significant work during the hackathon** – for example, use the event time to refine your model, explore new features, or at least rebuild parts of the solution in the official environment. This ensures the judges see fresh effort and innovation. If you leverage any pre-existing code, make sure it’s **openly available and credited**, and consider informing organizers or asking if such reuse is allowed. Many hackathons permit using open-source packages or boilerplate, but **simply re-submitting last year’s solution with minimal changes might be frowned upon**. In short, aim to treat your 2025-based pipeline as a starting point to accelerate development *during* the hackathon, rather than the final product itself.

2. Use of External Tools, Models, or Data

There is **no explicit ban on external libraries, tools, or datasets** in the provided rules. In fact, the hackathon workshop introduced participants to common machine learning packages like scikit-learn and TensorFlow on Day 1 ⁴, indicating that using standard open-source tools is not only allowed but expected. You are free to use any programming libraries or ML models (pre-trained or otherwise) that can help solve the problem. Similarly, we did not find any rule outright prohibiting external data – however, any **external resources must be properly integrated and documented** in your submission. The official instructions emphasize that **all custom code and any additional data files used must be included in your team’s repository** so that the solution is fully reproducible by judges ⁵. In other words, if you bring in an outside dataset or a pre-trained model, you need to **commit those files or provide code to fetch them** in your hackathon GitHub repo. Nothing should rely on an unavailable external service or data source at judging time.

While external data/tools are not banned, the competition's focus is clearly on the **provided dataset and problem**. The 2025 hackathon repository, for example, lists exactly which data files were given to teams (training data, test data, etc.) ⁶, implying that teams are expected to work primarily with those resources. Introducing additional datasets (for instance, public data on weather or regional factors) isn't mentioned in the rules, and doing so could be a double-edged sword. It might not be *discouraged* per se, but it could distract from the core problem or raise fairness questions if not every team has the same opportunity. If you believe an external dataset genuinely adds value, you **must include it in your repo and explain its use**, and be prepared to justify how it improves your model. Keep in mind that judges will scrutinize reproducibility – using exotic external data or services could complicate that. The safest approach is to **leverage the given data fully and the provided workflow tools** (the templates, baseline code, etc.) unless you have a very compelling reason to go beyond them.

Potential Conflict: Your current development leveraged the 2025 hackathon data and workflow, which was a smart way to prepare, but be cautious about carrying over any external assets into the 2026 competition. For example, if you integrated last year's data or a pre-trained model trained outside of the provided 2026 dataset, that might not align with competition expectations. Make sure that when the 2026 data is available, you train your models **solely on the official 2026 training set** (unless told otherwise). Any external code (utilities you wrote ahead of time, etc.) should be moved into your team's repo on the official hackathon GitHub and clearly documented. The judges will appreciate use of good tools – they explicitly **encourage standard ML practices and even provided a starter kit** – but they will disapprove if something isn't transparent or reproducible. In summary, you can use outside tools and even reference prior work, but **ground your solution in the hackathon-provided data and infrastructure**, and ensure everything needed to run your solution is checked in for the judges ⁵.

3. Data, Training, and Submission Format Restrictions

The hackathon comes with **strict submission requirements and formatting rules** that you must follow to avoid disqualification. Most importantly, the **output predictions file** (`solution.csv`) has a very specific format. **No deviations are allowed in this file's structure:** it must contain exactly the required columns and rows – nothing more, nothing less ⁷. Each well/fuel-type combination is one row (50 wells, with dual-fuel wells having two rows each, totaling 63 rows), and for each row you must provide a single point estimate and 100 uncertainty realization values named `Real_1` through `Real_100` ⁸ ⁹. Failing to include all 100 realizations **or** altering the file format (extra columns, missing entries, wrong file name, etc.) will **prevent your solution from being scored** by the automated system ¹⁰ ¹¹. The hackathon organizers made it clear that even a great project would “*not be scored*” if the submission format is wrong, which would be an avoidable disaster ¹¹. The good news is that your current pipeline already handles this: your context file notes that the solution CSV is produced in the exact format required (63 rows with all necessary columns) ⁹. Just be sure to **re-generate the file with the official 2026 test data** once it's provided, and double-check that the file name and structure match the template exactly (including spelling and capitalization of column names).

Aside from the prediction CSV, there are other **format and submission rules** to heed. Your **Jupyter notebook** containing the workflow must be submitted as well, and the filename **must be your team name** (e.g. `TeamName.ipynb`) ¹². The hackathon provided a template notebook (`Hackathon_ProjectTemplate.ipynb`) and expects teams to follow its section structure and naming conventions for consistency ¹³. Before final submission, ensure you have merged your code and analysis into the official template's format (sections for introduction, EDA, methodology, results, etc.) and then

rename the notebook file to exactly the required format (the instructions say “the file **must** be named `XXXX.ipynb` where XXXX is your team name”) ¹². **All team members’ names** should be listed in the notebook (often at the top or in a header cell) and especially on the first slide of your presentation. In fact, the rules specifically mandate that the first slide of the PowerPoint include the project title and everyone’s name – any omission was explicitly called out as something to avoid ¹⁴. The PowerPoint slide deck also needs to follow the provided template and be named `<TeamName>.pptx` before you push it to the repo ¹⁴ ¹⁵. Basically, **every deliverable has a naming convention and format**: `solution.csv`, `<Team>.ipynb`, `<Team>.pptx` are fixed requirements, and the content within must meet the guidelines (e.g. uncertainty included, all team members named, etc.). Missing a required file or misnaming it could get your team disqualified or penalized even if your analysis is excellent ¹¹.

Finally, note the **submission process and deadline**: All materials (code, data, notebook, slides, CSV) must be committed to the **official hackathon GitHub repository** provided to your team by the deadline (which in 2025 was noon the day after the hackathon, likely a similar timing for 2026) ¹⁶. Make sure you plan your work such that you can push final changes a bit before the cutoff – the automated scoring pipeline will pull your repo at the deadline, and late commits might be ignored. Given that you have been developing on a personal repo, you should integrate everything into the hackathon-provided repo as soon as you get it (at event start) and use that going forward. In short, **double-check every format and submission detail**: your current preparation covers the technical format well, but don’t forget to adjust file names and include all required documentation. A perfect solution can slip through the cracks if it doesn’t **follow the competition’s strict submission template** to the letter ¹¹.

Potential Conflict: Most of these rules align with what you’ve prepared, but there are a few things to adjust. For instance, you currently have your main notebook as `main_workflow.ipynb` – you will need to transfer its contents into the official template and rename it to your team name before submission ¹⁷. Also, ensure your presentation slides are created in the provided PowerPoint template (which you’ve saved as `Hackathon_PresentationTemplate.pptx` for now) and then saved as `YourTeamName.pptx` with all team members listed on slide 1 ¹⁵. If your repository contains any extra files or data not used in the final solution, remove or clearly organize them so as not to confuse judges. And of course, **do not forget to include the 100 uncertainty outputs** – your plan already has this covered with residual bootstrapping, but it’s worth emphasizing since forgetting this would “effectively nullify the submission” ¹⁰. In summary, pay meticulous attention to the submission format and repository contents, because any oversight there could invalidate your entry. Use the hackathon’s checklist (if provided) to verify you have: the correctly formatted CSV, the notebook in template form, the PPT, a README (optional but recommended), and that all are named properly and pushed to the GitHub on time.

4. Judging Criteria Emphasizing Originality and Compliance

The Energy AI Hackathon’s judging rubric is holistic – it’s not just about predictive accuracy, but also about **innovation, engineering quality, and communication** ¹⁸. According to the official criteria, judges will evaluate teams on multiple dimensions, including:

- **Technical Performance (Accuracy):** How well does your model predict the energy usage targets? An automated script will score your predictions against the ground truth after the event, so achieving low error is crucial ¹⁹. This is likely a significant portion of the score. Your current model (segmented random forests with cross-validation) should perform reasonably well here, but keep in mind others will be trying advanced techniques too. Ensure you validate your approach on the

training data (via cross-validation or holdouts) so you have confidence in its accuracy. If you have bandwidth during the hackathon, you might attempt some improvements (hyperparameter tuning, ensembles, etc.) to further reduce error – just remember that any improvement must be balanced against the time it takes and the risk of overfitting.

- **Solution Design & Implementation (Reproducibility and Code Quality):** Judges (and the hackathon “architects”) will **scrutinize your methodology and code**. The hackathon provided a Jupyter Notebook template and expects teams to follow that structure for clarity ¹³. They will look for a sound data science workflow: how you cleaned and explored the data, how you engineered features, which models you tried and why, and how you handled model training and uncertainty estimation ²⁰. Good practices like rigorous cross-validation, avoiding data leakage, and integrating domain knowledge are explicitly rewarded in this category ²⁰. Moreover, **code readability and documentation are critical** – your notebook should be well-organized with markdown explanations, and your code should be understandable to any competent engineer ²⁰. In 2025, the organizers even had an internal code review scoring sheet, so expect that messy or opaque code will cost points ²¹. For your project, this means you should take the time to **polish your notebook narrative**: include an introduction, clearly outline your approach, explain key decisions (e.g. why Random Forest? why split by fuel type?), and comment your code. If your current notebook is mostly code with minimal commentary, beef it up with markdown cells describing each step ²² ²³. The judges want to see that you didn’t just throw data at a black-box model – they want to see reasoning and reproducibility. Also double-check that anyone could rerun your notebook from scratch: if you have separate modules in `src/`, they should be imported properly, and any external data or requirements should be included. The fact that you’ve documented your environment (Python version, dependencies) and used a consistent project structure is a plus ²⁴. Just ensure when you migrate to the official repo that everything still runs smoothly from a clean start. In short, **make your code as clear and reproducible as possible**, following the given template, to maximize points in this area.

- **Innovation & Creativity:** Hackathon judges **explicitly reward originality** in the approach ²⁵. This category asks: *Did the team just apply a standard off-the-shelf method, or did they come up with fresh ideas and clever solutions?* The problem is open-ended, so there’s room to get creative. Teams that “inject original ideas (e.g., clever feature engineering, ensemble models, interactive visualizations, or domain-inspired features) tend to impress the judges” ²⁵. In your case, using separate models for each fuel type is actually a domain-inspired strategy – and it aligns well with the competition’s emphasis on incorporating oil & gas knowledge into the solution ²⁶ ²⁷. Be sure to **highlight that domain insight** in your presentation: for example, explain that you recognized Diesel, CNG, and Grid power usage behave differently, so you treated them separately. That demonstrates creative problem-solving grounded in domain expertise, which the judges appreciate (the challenge “explicitly encourages integrating domain expertise at every step” and rewards teams who do so ²⁶). That said, since you developed your solution based heavily on last year’s hackathon, it may resemble a “standard model” for this task (random forests were likely common in 2025). To stand out on innovation, consider what you can add or do differently this year. Perhaps you could try a new feature that wasn’t used before (are there interactions or non-obvious patterns you can capture?), or an improved uncertainty method, or even a simple visualization tool for the results. Even if you stick with the current model, maybe create an interactive plot or a well-thought-out analysis of feature importance – something to show extra effort in creativity. Remember that **innovation isn’t just about model choice**; it can also be in how you frame the problem, how you visualize outputs, or

how you integrate domain knowledge in a novel way. Ensure you articulate any such creative aspects in your submission. If your approach is essentially the same as a known baseline, the judges might see it as adequate but not “innovative.” So ask yourself: what’s one thing we can do or present that no other team is likely to have? Even a small unique insight can give you an edge in this category.

- **Presentation & Communication:** All the technical work culminates in your presentation to the judges on Sunday, and this is **a major factor in the scoring**. A polished, clear presentation can sometimes make the difference between a good project and a winning project ²⁸. The judges will be looking at how well you **explain your problem, approach, and results**. This includes how you **visualize the data and predictions** and convey the significance of your findings ²⁸ ²⁹. They have your slide deck in advance, and they may refer to it as you speak ³⁰ ³¹, so take care to make it self-explanatory and professional. The hackathon provided a presentation template, which you should use to organize your story (problem overview, methodology, results, conclusions) ³² ¹⁵. Make sure your slides are not overly crowded – use figures and bullet points for clarity – and **include key visuals**: for example, a plot of predicted vs. actual energy use for some wells, or a bar chart of feature importances, or a visualization of the uncertainty ranges. Judges *love* to see that you understand the data, so showing an exploratory analysis (trends, correlations, outliers) can earn you points for insight. In the judging of past hackathons, teams that clearly communicated their results and provided intuitive visualizations of model output were scored highly ²⁸ ²⁹. Conversely, a weakness in storytelling or visuals can hurt even a good project. From the project context, it looks like your current work may lack robust visualizations and exploratory analysis (this was noted as a gap) ³³. You should address that by generating some charts or graphs during the hackathon and incorporating them into your notebook and slides. Even simple plots (e.g., distribution of energy usage, or a comparison of model predictions vs. training data) will show the judges that you explored the data deeply. Finally, practice explaining your workflow in a concise narrative: you’ll likely have only ~10 minutes to present, so it needs to be clear and compelling. All team members should be introduced (and ideally each should speak if rules allow). Pay attention to the *story arc* – define the problem, describe your approach, highlight what’s novel, show results (with uncertainties), and conclude with the impact or next steps. A well-delivered presentation with a cohesive story can significantly boost your score, as evidenced by past events where presentation quality carried significant weight ²⁸.

- **Other Criteria (Impact and Completeness):** Some hackathons include a category for the potential impact or completeness of the solution. While the primary goal here is accurate energy forecasting, judges might give credit for solutions that **consider real-world deployment aspects** or future improvements ¹⁸. For instance, if you discuss how your model could scale to other datasets or how it might help industry decision-making, it shows you’re thinking beyond the competition. This isn’t as crucial as the core categories above, but if time permits, adding a slide or discussion on “Next steps / potential impact” could impress the judges as going above and beyond. Just ensure you’ve finished the required elements first (it’s worse to neglect the basics in favor of bonus points). Completeness also means you delivered all required pieces: correct CSV, notebook, and slides – which we covered in the submission rules section. Judges will notice if something is missing or incomplete, so don’t give them any easy reasons to dock points.

Recommendations: To maximize your scores and stay fully eligible, here are the key adjustments and emphases moving forward:

- **Develop on-site and document it:** Given that all coding should occur during the hackathon window, be prepared to **re-train your models and re-run analyses once the 2026 data is provided**, even if your pipeline is ready to go. Keep a log of what you do during the event (even commit messages can reflect incremental work) so that it's clear you didn't just copy in a finished product. Use the hackathon time to add improvements – for example, you might try an alternative algorithm or add a new feature interaction – and document this in your notebook as part of your development process. This will help satisfy any implicit expectation that you genuinely participated in the event's coding challenge, and it gives you fresh material to discuss in the presentation (“We tried X approach, but Y worked better...” shows effort and learning).
- **Stick to hackathon-provided resources:** Plan to use **only the official 2026 dataset for training and testing**. Do not incorporate last year's data into the training process for 2026 (unless the organizers explicitly encourage using past data, which they haven't). Everyone will be using the same new data, and that keeps the playing field level. You can, of course, use general knowledge gained from 2025 – for example, knowing which features tend to be important – but apply that knowledge by exploring the 2026 data anew. If you have external tools or references (say, a snippet of code from an open-source project you want to use), that's fine: just include the code in your repo and cite the source in a comment or your README. Ensuring **self-containment and transparency** of your project will keep you well within the rules ⁵.
- **Ensure format compliance:** Triple-check that your final outputs meet the hackathon's format requirements **exactly**. Before submitting, do a test run: clone your repo into a fresh environment and execute the notebook to produce `solution.csv`, then verify that file against the template (right number of rows, columns named correctly, etc.). Make sure your notebook filename is `YourTeamName.ipynb` and the slide deck is `YourTeamName.pptx` before the final git push ¹²¹⁴. Small details like including all team members on the first slide ³⁴ and in the notebook header can easily be overlooked in a rush – but they are explicitly required, so handle those early. It might help to create a short **submission checklist** for your team (e.g., “Rename notebook and slide files; names on slide 1; 100 realizations present; correct file names in repo; all data files included; requirements.txt updated; README updated; final push by XX:XX time”). This way you won't miss any compliance detail that could cost you points or eligibility.
- **Improve documentation and storytelling:** As noted, augment your Jupyter notebook with more markdown explanations and make sure it follows the template's section order ³⁵. Explain your reasoning at each step – this will satisfy the judges' criteria for clarity, and it doubles as preparation for your presentation narrative. Similarly, start drafting your slide content as you go, so you don't scramble Sunday morning. Include visuals: perhaps a flowchart of your workflow, graphs of some interesting data insights, and results. Address any currently missing pieces, such as exploratory data analysis or error analysis, to show the judges a comprehensive approach. If your current workflow skipped directly to modeling, take a step back during the hackathon to examine the data (even if you did so informally earlier, put it in the notebook now). For example, you could include a brief analysis of how energy usage varies by well characteristics or a sanity check of any outliers. This kind of **exploratory analysis and domain context** will strengthen your story and is often looked upon favorably by judges who want to see that you understand the “why” behind the numbers.

- **Emphasize innovation and originality:** Since you have a head start, use the extra time to push your solution a notch further in originality. Think about what other teams might do – many will train a model and provide uncertainties, but maybe you can add something extra. Perhaps implement a small interactive element in your presentation (if feasible), or try a quick alternative model (even if you ultimately stick with Random Forest, you could mention you tested a neural network or gradient boosting for comparison). Or you could calculate an interesting derived metric (e.g., “If our predictions were used, we estimate the company could save X% energy on average”). These are just ideas – the point is to have at least one **stand-out aspect** that you can point to as unique. Judges have a category for “Innovation,” and they will have seen many approaches; giving them something novel to remember helps. However, *do not compromise the core solution’s completeness* in pursuit of fancy extras. It’s better to have a solid, correct solution with a few nice twists than an over-ambitious idea that isn’t finished. Use your best judgment on what improvements yield the highest payoff in impressiveness relative to time spent.
- **Leverage domain knowledge and mentors:** The hackathon encourages integrating domain expertise ²⁶, so make sure you’re using the insights that an oil & gas context provides. Your fuel-type segmentation is a great example of this – clearly communicate that decision as a domain-driven improvement in accuracy ²⁷. If there are mentors available during the event, ask them for feedback on your approach or for any industry insights that could spark an idea (judges love when teams incorporate an insight gleaned from an industry mentor, as it shows collaboration and learning). Showing that you’ve considered the real-world context (why these predictions matter, how they might be used to optimize operations) can boost your **impact and completeness** scores. Even a brief mention in your presentation of how a field engineer might use your model’s output could set you apart as having a solution that’s not only accurate but practical.

In summary, **there is no red flag in the rules that outright disqualifies your current preparation** – you’ve been careful to follow the published submission format and used standard tools, which is good. The main areas to watch are the **timing/ethics of development** (ensure you do substantive work during the hackathon and don’t rely solely on pre-hackathon code) and the **judging expectations for novelty and communication**. The rules and judging criteria favor teams that produce original work on the spot, make full use of the provided data/tools, and present a polished, professional final product. By adjusting your plan to fit these guidelines – essentially, packaging your solution to meet every requirement and adding a dose of creativity in your analysis and presentation – you will ensure full eligibility and give yourselves the best shot at ranking highly. Good luck at the hackathon, and make sure to **finish strong with both compliance and creativity** in mind!

Sources:

- Energy AI Hackathon 2026 Judging Criteria and Submission Requirements ^{13 5 7 12 15 11 20 25 10 28}
- UT PGE Energy AI Hackathon Intro Slides and Info ^{1 4 2 3}
- Project Context and Analysis for Hackathon Solution ^{9 27 24 22 33 26}
- PGEHackathon 2025 Data Repository (problem description and templates) ⁶

1 Energy AI Hackathon - Hildebrand Department of Petroleum and Geosystems Engineering | The University of Texas at Austin

<https://www.pge.utexas.edu/energy-ai-hackathon/>

2 4 GitHub - PGEHackathon/workshop

<https://github.com/PGEHackathon/workshop>

3 5 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34

35 Energy AI Hackathon 2026 Judging Criteria and Submission Requirements.pdf

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6 GitHub - PGEHackathon/data2025

<https://github.com/PGEHackathon/data2025>